Example 2.54 Find the value of (i) $1+2+3+\ldots+50$ (ii) $16+17+18+\ldots+75$

Solution (i) $1+2+3+\cdots+50$

$$\text{Using, } 1+2+3+\cdots+n = \frac{n(n+1)}{2}$$

$$1+2+3+\cdots+50 = \frac{50\times(50+1)}{2} = 1275$$

(ii) $$16+17+18+\cdots+75 = (1+2+3+\cdots+75)-(1+2+3+\cdots+15)$$

$$= \frac{75(75+1)}{2} - \frac{15(15+1)}{2}$$

$$= 2850 - 120 = 2730$$



Progress Check

1. The sum of cubes of first n natural numbers is _____ of the first n natural numbers.
2. The average of first 100 natural numbers is _____.

Example 2.55 Find the sum of (i) $1+3+5+\cdots$ to 40 terms

(ii) $2+4+6+\cdots+80$ (iii) $1+3+5+\cdots+55$

Solution (i) $1+3+5+\cdots 40$ terms $= 40^2 = 1600$

(ii) $2+4+6+\cdots+80 = 2(1+2+3+\cdots+40) = 2\times\frac{40\times(40+1)}{2} = 1640$

(iii) $1+3+5+\cdots+55$

Here the number of terms is not given. Now we have to find the number of terms using the formula, $n = \frac{(l-a)}{d}+1 \Rightarrow n = \frac{(55-1)}{2}+1 = 28$

Therefore, $1+3+5+\cdots+55 = (28)^2 = 784$

Example 2.56 Find the sum of (i) $1^2+2^2+\cdots+19^2$

(ii) $5^2+10^2+15^2+\cdots+105^2$ (iii) $15^2+16^2+17^2+\cdots+28^2$

Solution (i) $$1^2+2^2+\cdots+19^2 = \frac{19\times(19+1)(2\times19+1)}{6} = \frac{19\times20\times39}{6} = 2470$$

(ii) $$5^2+10^2+15^2+\cdots+105^2 = 5^2(1^2+2^2+3^2+\cdots+21^2)$$

$$= 25\times\frac{21\times(21+1)(2\times21+1)}{6}$$

$$= \frac{25\times21\times22\times43}{6} = 82775$$

(iii) $$15^2+16^2+17^2+\cdots+28^2 = (1^2+2^2+3^2+\cdots+28^2)-(1^2+2^2+3^2+\cdots+14^2)$$

$$= \frac{28\times29\times57}{6} - \frac{14\times15\times29}{6} = 7714-1015 = 6699$$

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